

Exam 2

● Graded

Student

Cole West

Total Points

87 / 100 pts

Question 1

Question 1

3 / 3 pts

✓ - 0 pts Correct

Question 2

Question 2

3 / 3 pts

✓ - 0 pts Correct

Question 3

Question 3

0 / 3 pts

✓ - 3 pts Incorrect. The correct answer is (i).

Question 4

Question 4

3 / 3 pts

✓ - 0 pts Correct

Question 5

Question 5

0 / 4 pts

✓ - 4 pts Incorrect. The fully correct answer is both (ii) and (iv).

Question 6

Question 6

4 / 4 pts

✓ - 0 pts Correct

Question 7

Question 7

3 / 6 pts

✓ - 3 pts Work shown leads partway to the "clever" solution, but the process is not complete, nor does it lead directly to a conclusion (if one was drawn).

Question 8

Question 8

6 / 8 pts

8.1 Question 8(a)

6 / 6 pts

✓ - 0 pts Correct

8.2 Question 8(b)

0 / 2 pts

✓ - 2 pts Correct answer is ii)

Question 9

Question 9

14 / 14 pts

9.1 Question 9(a)

4 / 4 pts

✓ - 0 pts Correct

9.2 Question 9(b)

10 / 10 pts

✓ - 0 pts Correct

Question 10

Question 10

9 / 10 pts

✓ - 1 pt Sign mistake

Question 11

Question 11

8 / 8 pts

✓ - 0 pts Correct

Question 12

Question 12

12 / 12 pts

12.1 L

6 / 6 pts

✓ - 0 pts Correct

12.2 U

6 / 6 pts

✓ - 0 pts Correct

Question 13

Question 13

6 / 6 pts

✓ - 0 pts Correct

Question 14

Question 14

10 / 10 pts

14.1 Question 14(a)

6 / 6 pts

✓ - 0 pts Correct

14.2 Question 14(b)

4 / 4 pts

✓ - 0 pts Correct

Question 15

Question 15

6 / 6 pts

✓ - 0 pts Correct

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Student ID: 12580718

Math 3108 – Linear Algebra
Exam 2
Prof. Runnion
March 13, 2024

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has seven sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

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Multiple Choice and True/False. Fill in the box corresponding to the correct answer.

1. (3 pts) A basis for a vector space V is a _____ of vectors in V .
☐ i) minimal spanning set ☐ ii) maximal linearly independent set
☒ iii) both (a) and (b) ☐ iv) neither (a) nor (b)
2. (3 pts) For an $n \times n$ matrix A , if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then the columns of A are linearly independent.
☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
3. (3 pts) If there is an $n \times n$ matrix C such that $CA = I$, then A has n pivot positions.
☐ i) Always true ☒ ii) Sometimes true ☐ iii) Never true
4. (3 pts) If A and B are invertible $n \times n$ matrices, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
5. (4 pts) If the third row of the 4×4 matrix A is the sum of four times the first row and five times the second row, which of the following is true? Select all that apply.
☒ i) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
☐ ii) The column space of A is a subspace of $\mathbb{R}^4$.
☐ iii) The rows of A span $\mathbb{R}^4$.
☐ iv) The determinant of A is zero.
6. (4 pts) Let A be an $m \times n$ matrix. Which of the following is a subspace of $\mathbb{R}^m$? Select all that apply.
☒ i) $\text{Col } A$ (the column space of A)
☐ ii) $\text{Nul } A$ (the null space of A)
☐ iii) $\text{Row } A$ (the row space of A)

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Open Answer. Show ALL your work! Place your final answer in the box provided, if applicable.

7. (6 points) Show that the set $H = \left\{ \begin{bmatrix} a+2b \\ 3b \\ a-b \end{bmatrix} : a, b \text{ real} \right\}$ is closed with respect to scalar

multiplication. Note that this can be done the standard way or a "clever" way, both of which work quite nicely.

$$a \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

The set H is closed with respect to scalar multiplication because no rows contain a constant without a variable.

8. a) (6 points) Find a matrix A such that $W = \left\{ \begin{bmatrix} 5c-4d \\ 2c+d \\ c \end{bmatrix} : c, d \text{ real} \right\} = \text{Col } A$.

$$c \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}$$

$$W = \begin{bmatrix} 5 & -4 \\ 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}$$

$$A = \begin{bmatrix} 5 & -4 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Solution: } A = \begin{bmatrix} 5 & -4 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$$

- b) (2 points) The set W is a subspace of. (Fill in the box.)

☒ i) $\mathbb{R}^2$ ☐ ii) $\mathbb{R}^3$ ☐ iii) Neither of these.

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9. The matrices A and B given below are row equivalent. (You do *not* need to check this.)

$$A = \begin{bmatrix} 2 & -4 & -1 & -3 & 5 & 2 \\ -1 & 2 & -4 & -3 & -7 & -1 \\ -1 & 2 & -13 & -12 & -16 & -1 \\ 5 & -10 & 5 & -10 & 20 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- a) (4 points) Find a basis for the column space of A .

$$\text{basis col } A = \left\{ \begin{bmatrix} 2 \\ -1 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ -7 \\ -3 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -3 \\ -12 \\ -16 \end{bmatrix} \right\}$$

- b) (10 points) Find a basis for the null space of A .

$$x_1 = 2x_2 - 3x_5 - x_6$$

$$x_3 = -x_5$$

$$x_4 = 0$$

$$x_2, x_5, x_6 \text{ are free}$$

$$x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{basis for } N(A) = \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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10. (10 points) Consider the system of equations
$$\begin{cases} x + 2z = 3 \\ -2x + y + z = 4 \\ x - 3y + z = 1 \end{cases}$$

Given that $\begin{vmatrix} 1 & 0 & 2 \\ -2 & 1 & 1 \\ 1 & -3 & 1 \end{vmatrix} = 14$, use Cramer's Rule to solve for the variable x only.

$$\det A = 14$$

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -2 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

$$x = \frac{\det A_x}{\det A}$$

$$x = \frac{14}{14} = 1$$

$$\begin{aligned} \det A_x &= \begin{vmatrix} 3 & 0 & 2 \\ 4 & 1 & 1 \\ 1 & -3 & 1 \end{vmatrix} = 3 \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix} - 0 + 2 \begin{vmatrix} 4 & 1 \\ 1 & -3 \end{vmatrix} \\ &= 3(1 - (-2)) + 2(-12 - 1) \\ &= 3(3) + 2(-13) \\ &= 9 - 26 \\ &= -17 \end{aligned}$$

Solution: $x =$

$$\boxed{1}$$

11. (8 points) Assume that the matrices shown below are partitioned conformably for block multiplication. Compute the product.

$$\begin{bmatrix} I & 0 \\ -X & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I(A) + 0(C) & I(B) + 0(D) \\ -X(A) + I(C) & -X(B) + I(D) \end{bmatrix}$$

$$= \begin{bmatrix} A & B \\ -XA + C & -XB + D \end{bmatrix}$$

Solution:

$$\begin{bmatrix} A & B \\ -XA + C & -XB + D \end{bmatrix}$$

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12. (12 points) Find the LU factorization of the matrix $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$.

$$\begin{array}{l} R_1 + R_2 \rightarrow R_2 \\ -2R_1 + R_3 \rightarrow R_3 \end{array} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix} = U$$

$$\text{Solution: } L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

13. (6 points) Show that the set of all polynomials of the form $\mathbf{p}(t) = 1 + bt^2$ where a and b are real is *not* closed with respect to vector addition.

$$\begin{aligned} q &= 1 + xt^2 \\ p &= 1 + ut^2 \\ x \text{ and } u \text{ are real} \end{aligned} \quad \begin{aligned} q + p &= 1 + xt^2 + 1 + ut^2 \\ &= 2 + (x+u)t^2 \neq 1 + bt^2 \end{aligned}$$

$\therefore \mathbf{p}(t) = 1 + bt^2$ is not closed under vector addition
because $2 + (x+u)t^2 \neq 1 + bt^2$.

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14. a) (6 points) Use the method of your choice to calculate A^{-1} given $A = \begin{bmatrix} -2 & -3 \\ 2 & 1 \end{bmatrix}$.

$$\det A = (-2)(1) - (-3)(2) \\ = -2 + 6 \\ = 4$$

$$\frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = A^{-1}$$

$$\frac{1}{4} \begin{bmatrix} 1 & 3 \\ -2 & -2 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

Solution: $A^{-1} = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

- b) (4 points) Use the inverse you found in part (a) to solve $A\mathbf{x} = \mathbf{b}$ given $\mathbf{b} = \begin{bmatrix} -8 \\ 4 \end{bmatrix}$.

$$A^{-1}\vec{b} = \mathbf{x}$$

$$\begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} -8 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{4}(-8) + \frac{3}{4}(4) \\ -\frac{1}{2}(-8) + (-\frac{1}{2})(4) \end{bmatrix} = \begin{bmatrix} -2 + 3 \\ 4 + (-2) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Solution: $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

15. (6 points) Find the area of the parallelogram whose vertices are $(0,0)$, $(-2,3)$, $(3,3)$, and $(5,0)$. Place your answer in the box provided.

↳ no transformations needed

$$\begin{vmatrix} -2 & 3 \\ 3 & 3 \end{vmatrix} = -2(3) - (3)(3) \\ = -6 - 9 = -15 \Rightarrow 15$$

$$\text{Area} = |\det|$$

Area = 15